Building and Configuring Zigbee2MQTT with BLZ Protocol Support

version: 2.0

There are three ways to run Zigbee2MQTT with BLZ support:

1. **Home Assistant Add-on** (easiest, for HAOS users)
2. **Docker** (for standalone servers)
3. **From source** (for development)

Option 1: Home Assistant Add-on (HAOS)

Prerequisites

- Home Assistant OS with Supervisor
- MQTT broker (install the Mosquitto add-on from the HA Add-on Store)

Steps

1. Go to **Settings** > **Add-ons** > **Add-on Store** (three-dot menu)
2. Click **Repositories** and add: `https://github.com/fangzheli/hassio-zigbee2mqtt`
3. Refresh, find **Zigbee2MQTT Edge - BLZ** and click **Install**
4. In the add-on Configuration tab, set:
5. Serial port: `/dev/ttyUSB0` (or your dongle path)
6. Adapter: `blz`
7. Baud rate: `2000000`
8. Start the add-on

Option 2: Docker

```
docker run -d \
  --name zigbee2mqtt-blz \
  --restart=unless-stopped \
  --device=/dev/ttyUSB0:/dev/ttyUSB0 \
  -p 8099:8099 \
  -v $(pwd)/z2m-data:/app/data \
  -e TZ=UTC \
  ghcr.io/fangzheli/zigbee2mqtt:latest-dev
```

Create `z2m-data/configuration.yaml`:

```
homeassistant:
  enabled: false
frontend:
  enabled: true
  port: 8099
mqtt:
  base_topic: zigbee2mqtt
  server: mqtt://your-mqtt-broker:1883
serial:
  port: /dev/ttyUSB0
  adapter: blz
  baudrate: 2000000
```

Option 3: From Source

Prerequisites

- An MQTT Broker (e.g., Mosquitto) installed and running
- Git and Node.js (v20+) with pnpm installed

- For an overview of a standard setup, see the official Zigbee2MQTT Installation guide

Step 1: Install Local Dependency (zigbee-herdsman)

```
git clone -b feat/blz https://github.com/fangzheli/zigbee-herdsman.git
cd zigbee-herdsman
pnpm install
cd ..
```

Step 2: Clone and Build Zigbee2MQTT

```
git clone -b feat/blz-local-dev https://github.com/fangzheli/zigbee2mqtt.git
cd zigbee2mqtt
pnpm install
```

Step 3: Configure Zigbee2MQTT for BLZ

Open (or create) `data/configuration.yaml`:

```
mqtt:
  base_topic: zigbee2mqtt
  server: mqtt://localhost:1883
serial:
  port: /dev/ttyUSB0
  adapter: blz
  baudrate: 2000000
frontend:
  enabled: true
  port: 8099
```

Step 4: Start Zigbee2MQTT

```
pnpm start
```

Additional Notes

- BLZ adapter uses **2 Mbaud** serial communication
- Ensure your MQTT Broker is running before starting Zigbee2MQTT
- The BLZ dongle firmware must match the protocol version supported by zigbee-herdsman
- For firmware flashing, see the ThirdReality USB Dongle Guide